Project title: Design and Development of a Portable Rechargeable Digital Height Measuring Device Using Arduino and Ultrasonic Sensor

The Portable Digital Height Measuring Device is an electronic system designed to measure a person's height using an Arduino microcontroller, an ultrasonic sensor, and a digital display. The system measures the distance between the sensor and the top of the person's head and calculates height using a known reference height.

The device includes a display for real-time measurement, a rechargeable battery with USB charging, and an ON/OFF button for convenient operation. Its compact enclosure makes it suitable for use in schools, healthcare camps, homes, and educational laboratories.

Problem Statement

Conventional height measurement generally requires a measuring tape or wall-mounted scale and manual reading. This can introduce reading errors and may require a fixed measurement location.

The objective of this project is to develop a portable, rechargeable, and easy-to-use electronic device that measures height digitally, displays the result clearly, and reduces dependence on manual measurement.

Key Features

1. Touchless height measurement
2. Measures height in cm / foot and inches.
3. Displays the results on LCD screen
4. Provides USB charging and rechargeable battery operations
5. Include a physical ON/OFF button.
6. Design a compact, Easy to install, portable enclosure.

Future Features

1. Wi-fi connectivity data shareable to app
2. App will calculate BMI / suggest or plan diet, exercise
3. Scan ID card , details of the users will be shown , height calculated then store the height in the user profile
4. Voice Output
5. Calibration and Settings Menu

Hardware Requirements

1. Arduino Nano or Arduino Uno

Main controller

Processes sensor readings, calculates height, and controls the display. Arduino Nano is a convenient choice for a compact device.

2. HC-SR04 Ultrasonic Sensor

Measures the distance to the person's head by transmitting ultrasonic pulses and measuring their echo time.

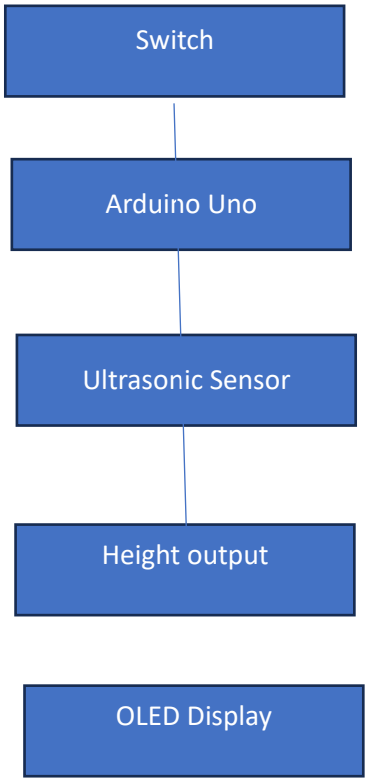
3. OLED or I²C LCD Display

Shows the calculated height, measurement status, and error messages.

4. ON/OFF Switch

Disconnects the power supply when the device is not in use.

Block Diagram



Working Principle

1. The HC-SR04 sensor sends an ultrasonic pulse towards the person's head.

2. The pulse reflects from the target and returns to the sensor.

3. The Arduino measures the time taken by the pulse to return.

4. The distance between the sensor and the head is calculated.

5. The Arduino subtracts this distance from the known sensor height above the floor.

6. The resulting height is displayed in centimetres.

Advantages

1. Portable and rechargeable design.

2. Digital display with automatic calculation.
3. Reduced manual reading and calculation.
4. Affordable components and straightforward programming.
5. Expandable architecture for smart features.
6. Suitable for educational and embedded-systems demonstrations.

Limitations

1. Measurement depends on a stable, known reference height.
2. Hair, head position, sensor angle, and sound reflections can affect readings.
3. Battery runtime depends on actual current consumption.
4. The basic design does not store measurement history.
5. The device must be calibrated and experimentally tested before being relied upon for health-related measurements.

Conclusion

The Arduino-based Portable Digital Height Measuring Device combines ultrasonic distance measurement, embedded programming and digital display technology in a compact system.

The basic prototype provides digital height calculation, while future additions such as Bluetooth connectivity, IoT monitoring, data storage, voice output, and battery monitoring can transform it into a smart measurement platform.